T1 Pencil contacts

Can pencil make copper tape's glue conduct? Build on 160–200 gsm card.

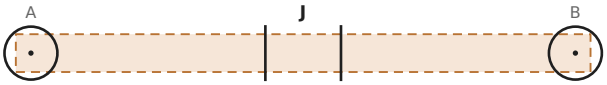
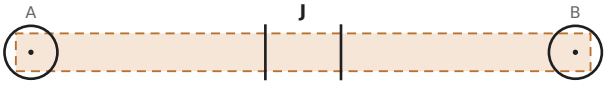
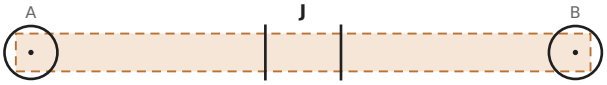
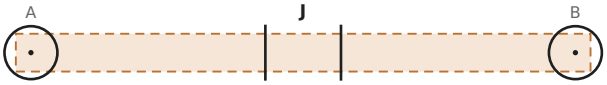
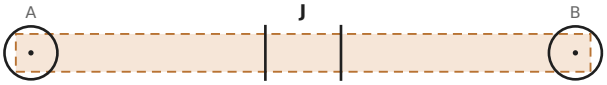
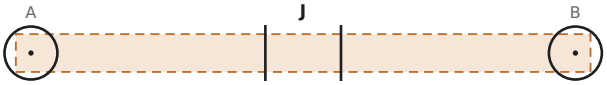
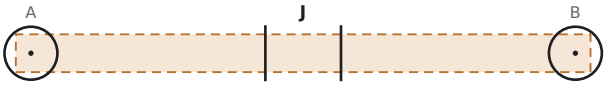
H1 BUS A · NC

1 Tape overlaps

Each lane is two pieces of tape overlapping by 10 mm at J. Before you lay the right-hand piece, treat the glue on its first 10 mm as labelled. Measure A to B in ohms.

Meter zero _____ Ω **Built on** _____

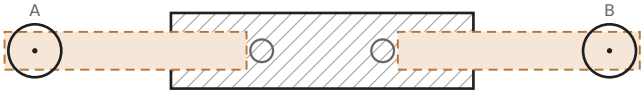
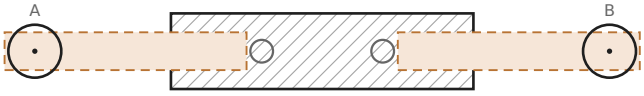
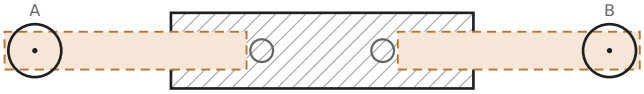
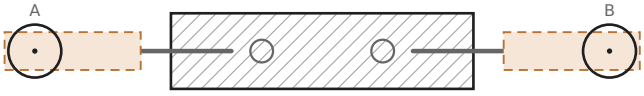
Paper _____

		Day 0	Flexed	Day 7
O0 Continuous the reference				
O1 Plain overlap control: glue untouched				
O2 Scratched fingernail through the glue				
O3 Pencil rubbed 6B rubbed into the glue				
O4 Scratched + pencil scratch, then rub 6B in				
O5 Scratched + HB harder tip, less graphite				
O6 Sticky to sticky both glue faces rubbed with 6B				
O7 Your version				

H2 V+ · NC

2 Tape on pencil

Shade each track with 6B, then join copper to both ends as labelled. Measure A to B, then the bare pencil between the two small circles. Readings are in kΩ.

		A to B	Pencil	Day 7
K1 Plain tape on pencil control: glue on the pencil				
K2 Scratched + pencil treat the glue, as O4				
K3 Folded tab copper faces down, clear tape				
K4 Bare wire wire onto pencil, clear tape				
K5 Your version				

H3 GND · NC

Notes

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.

T1 Running the pencil tests

Print this sheet once on plain paper. Print sheet 1 on card, once per paper you compare.

H1 BUS A · NC

You need

- 5 mm copper tape and clear tape
- A 6B and an HB pencil, sharpened, and an eraser
- A few centimetres of bare wire, or cut-off resistor legs, for K4
- A multimeter with 200 Ω and 200 kΩ ranges
- Your T0 results: O1 should behave like T0's J2

Before you start

1. Print sheet 1 at 100 % on card. Write the paper on it: brand, weight and finish.
2. Zero the meter on its 200 Ω range and write the reading on sheet 1.
3. Keep fingers off the glue except where a step says to treat it: skin oil changes results.

1 · Tape overlaps

1. Lay the left-hand piece of each lane from A to just past J.
2. Cut the right-hand piece and peel its backing. Treat the glue on its first 10 mm: **scratched** means a criss-cross scored with a fingernail; **pencil** means rubbing the tip back and forth until the glue looks grey.

H2 V+ · NC

3. Lay it over the left-hand piece by 10 mm, from J to B, and burnish the overlap hard with a fingernail.
4. **O6, sticky to sticky:** fold the left-hand piece's last 10 mm back over itself so its glue faces up. Rub 6B into that glue and into the right-hand piece's glue, then lay the right-hand piece on top.
5. Brush loose graphite off the paper. A smudge beside a lane is a leak.
6. Measure A to B on the 200 Ω range. **Flexed** and **Day 7** work as on T0.

Trying different papers

Paper changes how pencil takes, how well tape grips and how flaps spring back. Build sheet 1 on each paper you're considering and compare:

- Smooth card, 160–200 gsm: the standard's choice
- Heavier card, 250–300 gsm, if your printer takes it
- Cartridge or drawing paper, 120–160 gsm: more tooth for pencil
- Coated (silk or gloss) card: tape grips well, pencil may not
- Plain 80 gsm paper, as a baseline

H3 GND · NC

The pencil-only reading compares how each paper takes graphite: lower is better. Overlays are a separate question: try printing on vellum or tracing paper too, as inkjet may smudge.

2 · Tape on pencil

1. Shade each track with the 6B: at least 10 passes, until it shines.
2. Join copper to both ends of each track. **K1:** lay the tape straight on. **K2:** scratch and rub 6B into the glue's end first. **K3:** fold the tape's last 10 mm back on itself, sticky to sticky, so copper faces down; hold it on the pencil with clear tape. **K4:** stop the tape short, and bridge to the pencil with bare wire under clear tape. **K5:** your idea.
3. Switch to the 200 kΩ range. Measure A to B, then the bare pencil between the two small circles.
4. **Day 7:** measure A to B again.

Pass or fail

- **Overlaps:** pass if within 1 Ω of O0 and steady. If O3 to O6 pass where O1 fails, pencil-treated overlaps could replace solder.
- **On pencil:** a good join makes A to B only a little more than the pencil-only reading. Several times more, OL, or a wandering reading means a poor contact.

Results for standard v0.2

Pencil overlaps OK?	
Best overlap treatment	
Best join to pencil	
Best paper	

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.